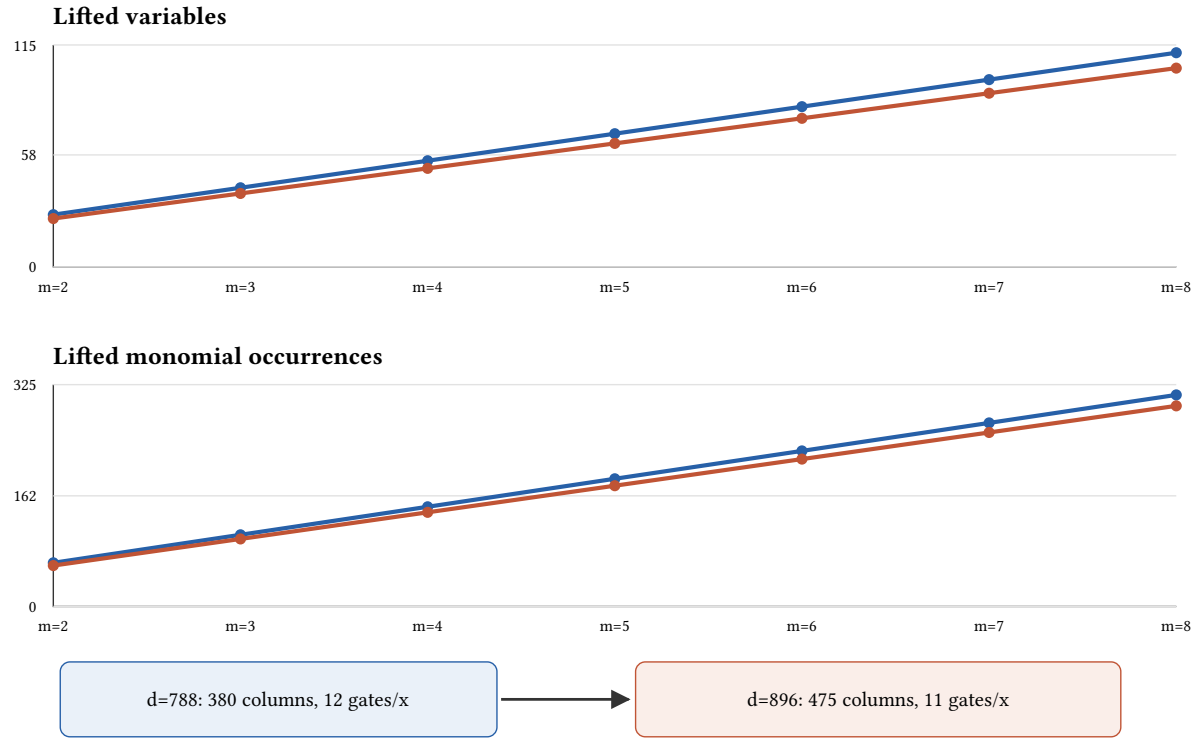
Exact m=2…8 inputs for the selected 896-coset base

Frozen algebraic inputs and independent planted-equation replay, 2026-10-17 UTC. The selected target-blind base has 475 points on the certified 63-bit prime-order curve. The new record serializes direct and quadratic-lifted chained-S3 systems for every m=2…8. These are correctness inputs, not observed solver costs or natural relations.

Prime coset systems: m=2…8 exact input counts

Same 63-bit curve; d=788 has 380 columns, d=896 has 475. No solver costs measured.



All lifted systems have maximum input degree 4. Base sizes differ; counts do not predict Gröbner degree or speed.

d=788 SHA-256: 4ed209221a30fcfe554a2adb8b782c77082910a37e13513cbe25bc7beecfc731

d=896 SHA-256: e5bc89daa181eac56e5a854895bfb366e5ffe3ca4d83b9e04930f232251ff72f

At m=8 the d=896 lifted input has 103 variables, 104 equations and 294 monomial occurrences, versus 111, 112 and 310 for the prior d=788 input. The new base has 475 rather than 380 columns. The d=896 square-and-multiply chain uses 11 gates per x instead of 12; both lifted systems have maximum input degree four. Base sizes differ, and input counts predict neither Gröbner degree nor runtime.

Independent check

SageMath 10.10.rc0 checked all seven known-point sums, 70 direct and 455 lifted equations, and an altered-auxiliary rejection at each arity. The 896-element coset has 421 x-values that do not lift to the curve; arbitrary solver roots still need point and sign checks. Direct S8 addresses seven summands plus target; eight summands need S9 or a chained construction.

Remaining gate

Freeze equal-realized-size 475-point random and small-x controls, then run bounded matched solver comparisons under an approved protocol. Natural relations, useful rank, unseen target work, complete setup and memory charges, and matched rho remain unmeasured. The companion Markdown report links the exact systems and independent replay.